

School mapping in ESRI imagery

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Recap from last meeting

Model fine-tuning

- All Anditi schools + equal number of OSM Vietnam non-schools
- Static Gaussian non-school sampling
- 20 epochs
- Train results
 - **F1: 99.55%**
 - P: 99.80%
 - R: 99.30%

Dense inference

- Use fine-tuned model on **overlapping satellite images** (tiles) from chosen districts
- Perform non-maximum suppression (**NMS**) to reduce the number of duplicate detections
- Measure **number of detected Anditi schools**
- Deliver **list of potential schools** sorted by prediction confidence

Dense inference results (after NMS)

- Found 900/946 (**95.14%**) Anditi schools
- Recall (min probs): 95.14%
- Recall (mean probs): 95.14%
- Recall (max probs): 95.14%
- 46 not detected schools (**4.86%**)
- List of potential schools: **9573 images**

Problems

- No dataset for evaluating the final fine-tuned model
- Only measured recall, not precision and **F1 score**

New contributions

New validation datasets

- Two hand-collected datasets
- Can be used for evaluating the final fine-tuned model

Exhaustive (narrow) dataset

- Locations from 4 wards
- **110 schools** (exhaustive search)
- **1163 overlapping tiles** total (after filtering)
- Can be used for calculating precision, recall and **F1 score**

Wide dataset

- Locations from 5 districts
- **93 schools** (not exhaustive search!)
- **20 981 overlapping tiles** total (after filtering)
- Can be used for calculating recall

Threshold validation

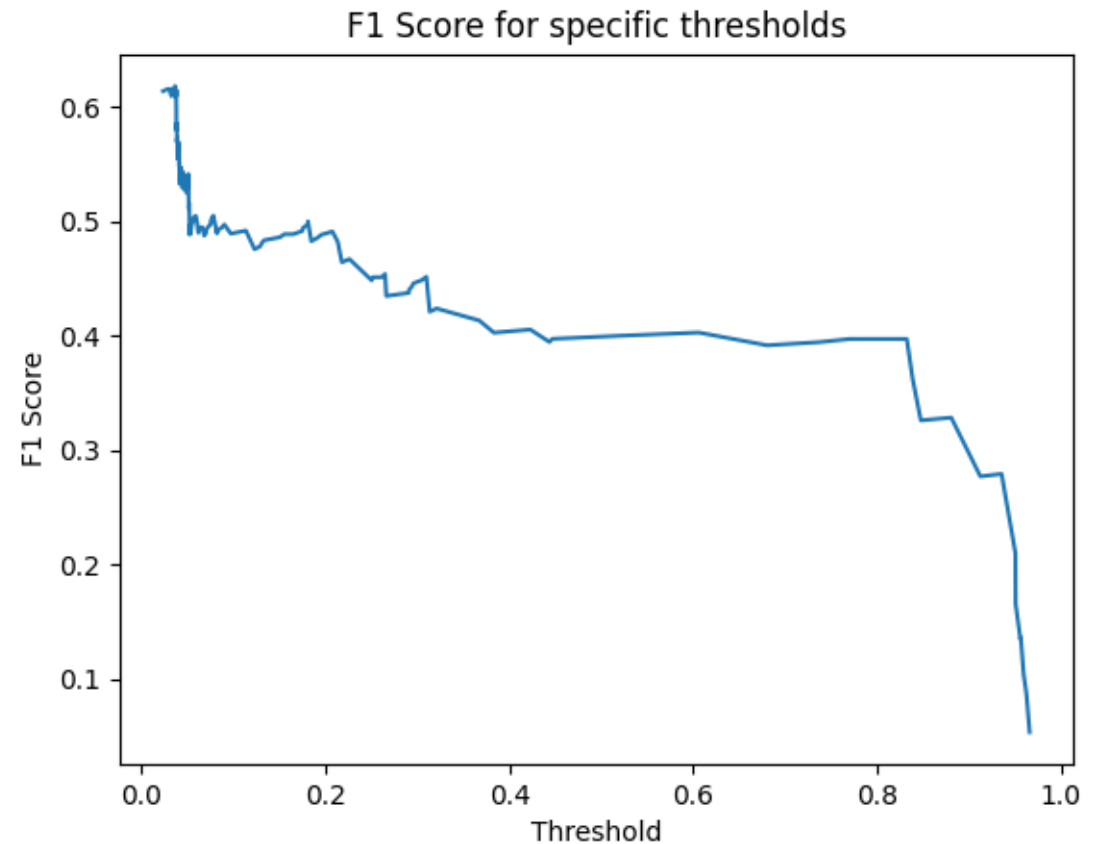
- We want to determine the ideal classification threshold
- Get dense inference predictions for exhaustive dataset
- Measure F1 score on predictions for variable thresholds (0-1)
- Pick threshold with the highest F1 score
- Get recall, precision and F1 score for chosen threshold

Threshold validation, 1:1 model

- Fine-tuned for **20 epochs**
- **Dynamic Gaussian** non-school sampling
- School : non-school ratio – **1 : 1**

Threshold validation, 1:1 model

- **Final threshold: 3.61%**
- Final precision: 45.34%
- Final recall: 97.27%
- **Final F1 score: 61.85%**

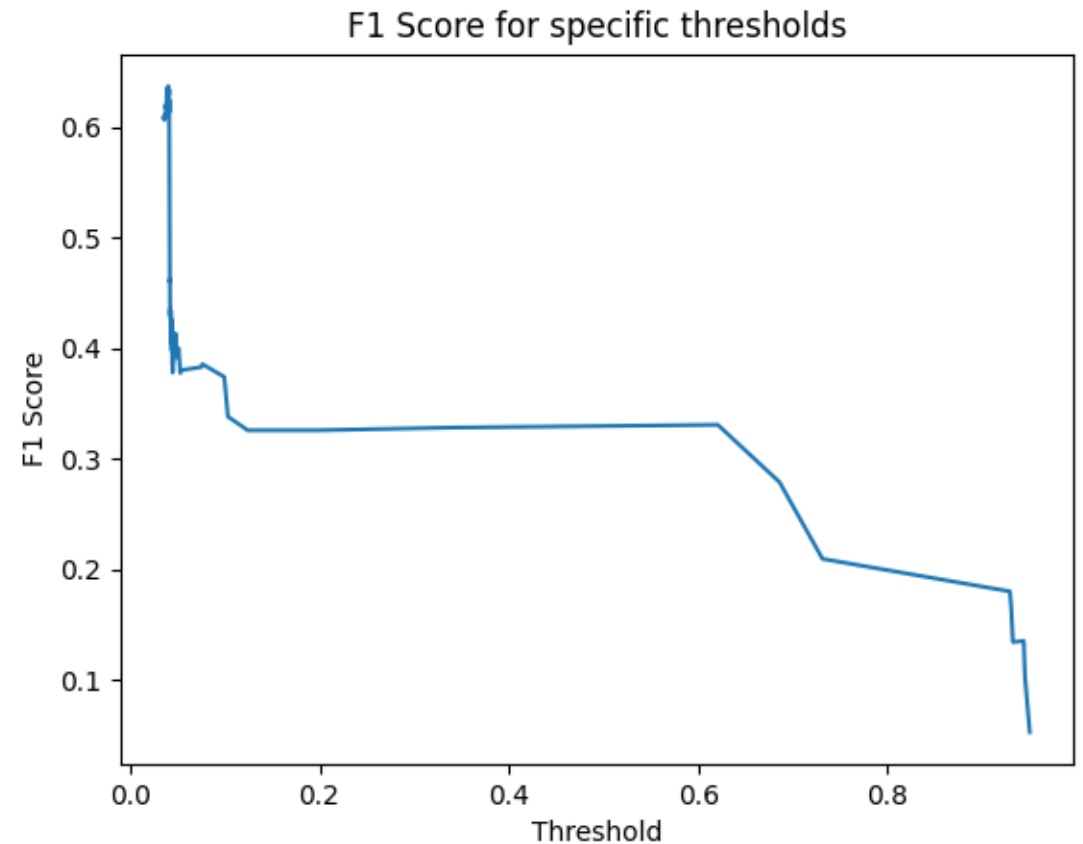


Threshold validation, 1:10 model

- Fine-tuned for **20 epochs**
- **Dynamic Gaussian** non-school sampling
- School : non-school ratio – **1 : 10**

Threshold validation, 1:10 model

- **Final threshold: 4.04%**
- Final precision: 51.10%
- Final recall: 84.55%
- **Final F1 score: 63.70%**

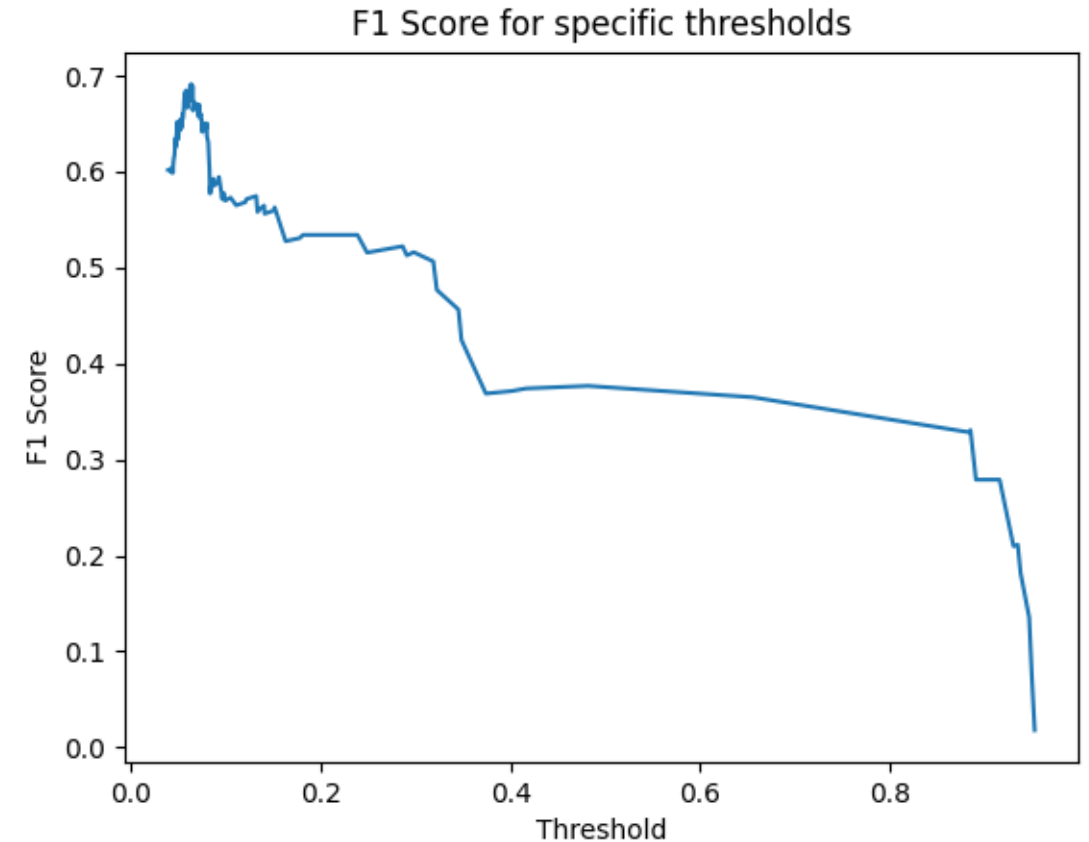


Threshold validation, 1:10 model, 1 epoch

- Fine-tuned for **1 epoch**
- **Dynamic Gaussian** non-school sampling
- School : non-school ratio – **1 : 10**

Threshold validation, 1:10 model, 1 epoch

- **Final threshold: 6.33%**
- Final precision: 63.16%
- Final recall: 76.36%
- **Final F1 score: 69.14%**



Conclusion & Future work

- Fine-tuning for 20 epochs may overfit the model to Anditi dataset
- Explore and test options for reducing the overfitting of our model
- Validate newly fine-tuned model on exhaustive, narrow and Anditi dataset